

Torque ripple minimization of PMSM using resonant controller in robot applications

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Abstract - Torque ripples in Permanent Magnet Synchronous Motors (PMSMs) are a significant challenge in high-precision applications, as it limits servo control accuracy and speed stability. A novel scheme is proposed to minimize torque ripples by a resonant controller based on speed harmonic minimization in this paper. This scheme leverages the relationship between speed harmonics and injected voltage harmonics, where the voltage harmonics are directly generated by the speed resonant controller responding to the detected speed harmonics. Compared to state-of-the-art methods for reducing torque ripple, the proposed scheme is simpler to implement in robotic systems, as it abandoned current harmonic compensation link and does not depend on specific motor parameters or motor control parameters. It has been simulated and experimentally validated under two test benches. The results demonstrate that the proposed scheme enhances servo system control accuracy.

Index Terms - torque ripple compensation, speed harmonic minimization, resonant controller, PMSM.

I. INTRODUCTION

As a critical active component in robot drivetrains, torque ripple in PMSMs can significantly limit servo control accuracy and speed stability, particularly in high-precision robotic applications. For example, torque ripple can lead to acoustic noise, mechanical vibrations, and reduced robot accuracy [1]-[3]. As a control strategy for torque ripple reduction, current harmonic injection compensation is primarily used to reduce torque ripple. It can be further divided into two approaches: direct minimization of current harmonics [4]-[7] and speed-based harmonic compensation. [8]-[13].

Methods for the direct minimization of current harmonics typically involve building a torque ripple model combined with various algorithms. To accurately estimate the torque ripple in PMSMs, factors such as current harmonics, PM flux harmonics, cogging torque [4][5], and inductance harmonics [6] are considered. Based on these models, different algorithms are designed to minimize torque ripple, including the genetic algorithm [4], back EMF-based compensation [5], and artificial neural networks [7]. However, the accuracy and effectiveness of current harmonic compensation are directly influenced by the complexity of the torque ripple model. These methods heavily depend on machine parameters and less effective in addressing high-frequency torque ripples.

Speed-based methods, a type of feedback compensation technique, focus on reducing speed harmonics to minimize

torque ripple. The key principle of these methods is generating a superimposed current to counteract undesired torque fluctuations. By producing an optimal current harmonic, speed harmonics can be reduced. Since the magnitude of a speed harmonic is proportional to the corresponding torque harmonic [8], different algorithms are applied to minimize speed harmonics. In [9]- [11], proportional-integral-resonant (PIR) controllers are designed to analyze the impact of torque ripple on speed and current. These compensation approaches do not require motor parameter dependency, making these methods more adaptable to various operating environments. However, a current controller is required, which generates current harmonics from speed harmonic. Then a voltage harmonic controller, which generates the corresponding voltage into motor system. The involvement of current harmonic control as an intermediate step increases system complexity and potential instability.

To address this issue, Wu et al. [12] propose a voltage harmonic compensation method based on a Gradient Descent Optimization (GDO) algorithm. By analyzing the relationship between speed harmonics and injected voltage harmonics, the method utilizes q-axis injected voltage harmonics directly to adjust the amplitude of speed harmonics. This approach is simpler to design and implement. However, since the GDO algorithm is used to optimize the injected voltage harmonic defining the appropriate conditions to obtain the optimal values remains a challenge. The optimization process may introduce uncertainty in achieving the desired results. Additionally, the accuracy of speed harmonic extraction directly affects the overall effectiveness of speed harmonic compensation.

Due to the highly dynamic and nonlinear nature of robot manipulators, this article proposes a voltage harmonic compensation method to minimize torque ripple in SPMSMs. The method directly injects voltage harmonics into the motor system based on speed harmonics using a resonant controller. Unlike traditional approaches, it eliminates the need for a current harmonic compensation link, simplifying the system. The resonant controller, designed using a second-order low-pass filter, allows independent control of the cutoff frequency and resonance. This approach is easy to implement, overcomes the uncertainty induced by algorithms, and offers flexibility in handling variable frequencies and resonance for effective torque ripple compensation.

The paper is structured as follows: Section II describes the torque ripple model of the PMSM. Section III introduces the proposed scheme, which is designed based on a speed harmonic resonant controller. Section IV presents the simulation results, while Section V discusses experimental testing and validation using two test benches. Finally, Section VI provides concluding remarks.

II. TORQUE RIPPLE MODEL OF PMSM

For the purpose of this article, torque ripple model (TRM) is discussed and analyzed. Considering harmonic components in PM flux linkage, cogging torque and stator currents, the electromagnetic torque of PMSM is expressed with variables in rotating reference frame (RRF) [9] as:

$$\begin{cases} \tau_r = 1.5p[(\psi_{sd}i_{sq} - \psi_{sq}i_{sd}) + \mathbf{i}_s(d\boldsymbol{\psi}_m/d\vartheta_e)] + \tau_{cog} \\ \boldsymbol{\psi}_s = \mathbf{L}_s \mathbf{i}_s + \boldsymbol{\psi}_m \end{cases} \quad (1)$$

where $\boldsymbol{\psi}_s = [\psi_{sd} \ \psi_{sq}]^T$, $\mathbf{i}_s = [i_{sd} \ i_{sq}]^T$, $\boldsymbol{\psi}_m = [\psi_{md} \ \psi_{mq}]^T$; ψ_{sd} and ψ_{sq} are stator flux components; i_{sd} and i_{sq} are stator current; ψ_{md} and ψ_{mq} are permanent magnet (PM) flux components; p is the number of pole pair; $\mathbf{L}_s = \text{diag}[L_{sd} \ L_{sq}]$ is the stator inductance matrix; ϑ_e is electrical angle; τ_{cog} represents the cogging torque. The matrix of stator currents and PM flux linkage can be expressed by the phase and amplitude in RRF, considering harmonics using the Fourier series as

$$\mathbf{i}_s = \begin{bmatrix} i_{sdF} \\ i_{sqF} \end{bmatrix} + \sum_{n=1}^{\infty} \begin{bmatrix} A_{idn} \cos(n\vartheta_e + \varphi_{idn}) \\ A_{iqn} \cos(n\vartheta_e + \varphi_{iqn}) \end{bmatrix} \quad (2)$$

$$\boldsymbol{\psi}_m = \begin{bmatrix} \psi_{mdF} \\ \psi_{mqF} \end{bmatrix} + \sum_{n=1}^{\infty} \begin{bmatrix} A_{\psi dn} \cos(n\vartheta_e + \varphi_{dn}) \\ A_{\psi qn} \cos(n\vartheta_e + \varphi_{qn}) \end{bmatrix} \quad (3)$$

$$\tau_{cog} = \sum_{n=1}^{\infty} [A_{\tau n} \cos(n\vartheta_e + \varphi_{\tau n})] \quad (4)$$

where i_{sdF} , i_{sqF} are current fundamental components; A_{idn} and A_{iqn} are magnitudes of harmonic currents; φ_{idn} and φ_{iqn} are phase of harmonic currents. ψ_{mdF} and ψ_{mqF} are fundamental components of PM flux; $A_{\psi dn}$ and $A_{\psi qn}$ are magnitudes of PM flux harmonics; φ_{dn} and φ_{qn} are phase of PM flux harmonics. n is the electrical harmonic order, ϑ_e is the electrical angle. The cogging torque τ_{cog} is also expressed as the Fourier series, where $A_{\tau n}$ and $\varphi_{\tau n}$ are magnitude and phase of the n th order cogging torque.

The electromagnetic torque of PMSM can be modelled as follow:

$$\tau_r = \tau_F + \tau_{rip} \quad (5)$$

where τ_r is the output torque of a PMSM, τ_F is the fundamental torque, τ_{rip} is the harmonic components. The d axis lies on the same axis of the magnets flux for SPMSMs, hence $\psi_{mqF} = 0$, $\psi_{mdF} = \psi_{pm}$, $L_{sd} = L_{sq}$. The fundamental and harmonic of torque can be expressed as:

$$\tau_F = 1.5p\psi_{pm}i_{sqF} \quad (6)$$

$$\tau_{rip} = \sum_{n=1}^{\infty} [T_n \cos(n\vartheta_e + \varphi_{\tau n})] \quad (7)$$

where T_n and $\varphi_{\tau n}$ are the amplitude and phase of torque ripple. The torque ripple is composed of nonlinear components via vary speeds which the model has been built in detailed model in [8][9]. Based on (1)-(7),

$$T_n = \sqrt{(1.5p)^2(E_n^2 - F_n^2) + A_n \cos \varphi_{\tau n} + A_n \sin \varphi_{\tau n}} \quad (8)$$

$$\varphi_{\tau n} = \arctan((A_{\tau n} \sin \varphi_{\tau n} + 1.5pE_n)/(A_{\tau n} \cos \varphi_{\tau n} - 1.5pF_n)) \quad (9)$$

where,

$$\begin{aligned} E_n &= n(i_{sdF}A_{\psi dn} \cos \varphi_{dn} + i_{sqF}A_{\psi qn} \cos \varphi_{qn}) \\ &\quad - i_{sdF}A_{\psi qn} \sin \varphi_{qn} + i_{sqF}A_{\psi dn} \sin \varphi_{dn} \\ &\quad + A_{iqn}\psi_{pm} \sin \varphi_{iqn} \\ F_n &= n(i_{sdF}A_{\psi dn} \sin \varphi_{dn} + i_{sqF}A_{\psi qn} \sin \varphi_{qn}) \\ &\quad + i_{sdF}A_{\psi qn} \cos \varphi_{qn} - i_{sqF}A_{\psi dn} \cos \varphi_{dn} \\ &\quad - A_{iqn}\psi_{pm} \cos \varphi_{iqn} \end{aligned} \quad (10)$$

III. TORQUE RIPPLE MINIMIZATION WITH PROPOSED SCHEME

A. Proposed Controller Design

From Equation (1), the torque ripple model of the PMSM is constructed to analyse the relationship between controlled and uncontrolled harmonics. The controlled harmonic originates from the current, while the uncontrolled harmonics include magnet flux harmonics and cogging torque. By injecting appropriate current harmonics, torque ripples caused by flux harmonics and cogging torque can be cancelled out. According to the relationship between speed harmonics and torque ripple [8], minimizing speed harmonics of the same order can effectively reduce torque ripple. Additionally, the current harmonic control link can be eliminated [12]. Ultimately, the problem of torque ripple compensation can be addressed by injecting suitable voltage harmonics to reduce speed harmonics at specific harmonic orders. Schematic of proposed method to minimize torque ripple in Fig. 1. There are three steps involved in generating additional voltage to minimize speed harmonics. The first step is harmonic extraction. The speed harmonic components are extracted from the measured speed signal. Then an additional voltage is generated based on the extracted harmonics using a harmonic controller. The last step is voltage compensation. The generated additional voltage is combined with the original voltage and fed into the motor control system, which operates based on the field-oriented control (FOC) strategy.

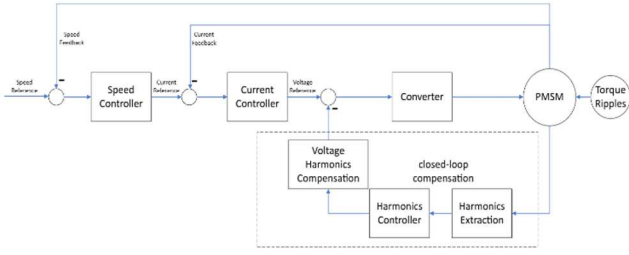


Fig. 1. Schematic of proposed method to minimize torque ripple

As a key component of the proposed method, the speed and error signals are used as inputs to the resonant controller. The controller extracts the n th-order speed harmonic from these signals and generates the corresponding voltage harmonic needed for compensation. This voltage harmonic is injected into the reference voltage generated by the current loop control. Since this method detects speed signals instead of current signals, it operates independently of current harmonics. As a result, the proposed method minimizes torque ripple not only for harmonic frequencies of the form $6n$, which correspond to current harmonics, but also for any n th electrical frequency order, offering greater harmonic frequency flexibility.

The speed which is a periodic disturbance can be informed of fundamental and harmonic, with speed harmonic can be described using a Fourier series in the form of a trigonometric series [11].

$$\omega_r = \omega_f + \omega_h = \omega_f + \sum_{n=1}^{\infty} [A_{\omega n} \cos(n\vartheta_e + \varphi_{\omega n})] \quad (11)$$

where ω_r is the mechanical rotary speed, ω_f is the fundamental component of the mechanical angular speed, ω_h is speed harmonic, $A_{\omega n}$ and $\varphi_{\omega n}$ are the amplitude and phase in n th electrical harmonic order. The block diagram of the proposed voltage harmonic control based on the resonant controller for SPMSM is shown in Fig. 2.

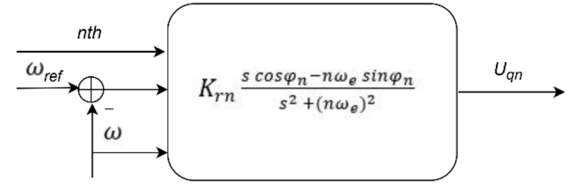


Fig. 2. The proposed resonant controller

For improving the dynamic characteristics and stability of motor, a phase margin is considered, and the transfer function of the resonant controller $G_{rc,n}$ is termed as follow [10][11].

$$G_{rc,n} = K_{rn} \frac{s \cos \varphi_n - n \omega_e \sin \varphi_n}{s^2 + (n \omega_e)^2} \quad (12)$$

where the harmonic orders n can be determined according to FFT analysis. K_{rn} is the resonant coefficient, and φ_n is a leading angle increasing phase stability margin with n th harmonic order of electrical frequency which are be set in the test [9]. ω_e is the electrical speed. The generated optimal voltage can be formed from the amplitude and cosine function with parameters the order of electrical angle and phase, as follows:

$$U = \sum_{n=1}^M A_n \cos(n\vartheta_e + \phi_n) \quad (13)$$

where U is the instantaneous value of the voltage harmonic, A_n is the n th order of amplitude of the voltage harmonic, M is the number of harmonics, and ϕ_n is the n th order of phase offset of the harmonic. The optimal voltages are injected into motor control system to minimize speed harmonics. Furthermore, the amplitudes and phases of optimal voltages can be obtained by extraction function [3].

B. Close-loop resonant based voltage harmonic control

By injecting the voltage harmonic into motor control based on speed harmonic, the block diagram of the proposed close-loop resonant based voltage harmonic control for PMSM is shown in Fig. 3.

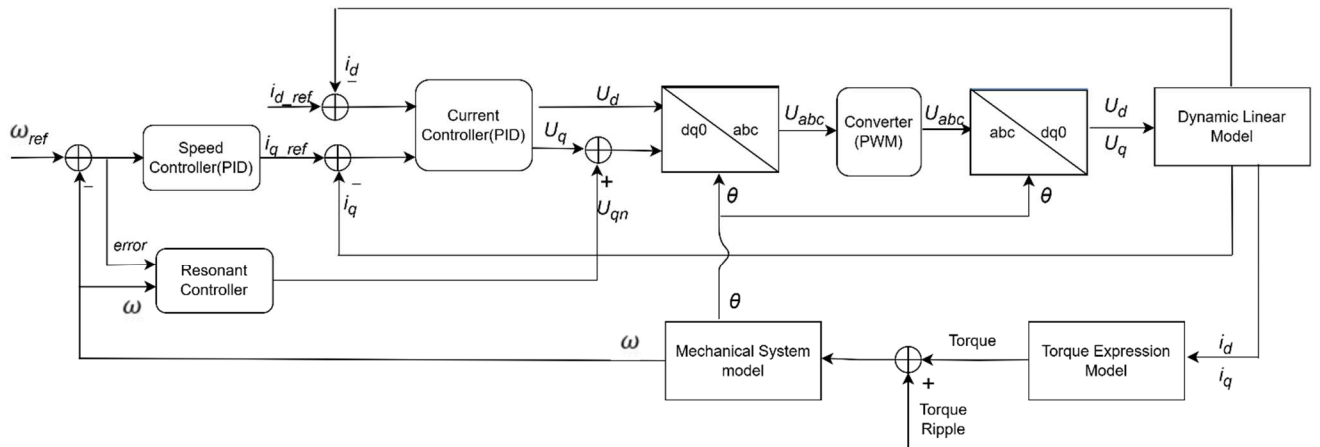


Fig. 3. Block diagram of the proposed close-loop resonant based voltage harmonic control for SPMSM

In SPMSM, the d -axis aligns with the magnet flux and does not contribute to torque ripple, whereas torque ripple is generated only on the q -axis. Therefore, the resonant-based voltage harmonic is injected exclusively into the q -axis. The compensated harmonic voltage, generated directly from the speed signal, serves as the output of the resonant controller. By adding this compensated voltage, speed harmonics of the same order are filtered, effectively reducing speed harmonics and even minimizing torque ripple. There are current and speed controllers with PI regulator to adjust motor control in Fig. 3. The transfer function can be expressed as

$$G_v(s) = K_{pv} + \frac{K_{iv}}{s} \quad (14)$$

$$G_{id}(s) = K_{pd} + \frac{K_{id}}{s} \quad (15)$$

$$G_{iq}(s) = K_{pq} + \frac{K_{iq}}{s} \quad (16)$$

where $G_v(s)$ is the transfer function of the speed loop control which contain n th speed harmonic, $G_{id}(s)$ and $G_{iq}(s)$ are the transfer functions of the current loop control on d and q axis with PI control.

C. Speed harmonic identification

From (11) and (12), the key is determining the electrical harmonic orders to the proposed algorithm. The experimental testing is conducted at a low speed of 10 rpm, making it easier to identify the speed harmonic order. The speed magnitude compares in time domain and FFT analysis are given as follow in Fig. 4.

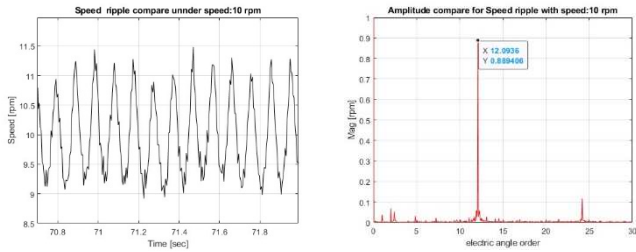


Fig. 4. Speed harmonic orders definition in time and frequency domain

From FFT analysis, in electrical angle condition at speed is 10 rpm, the total harmonic magnitude is about 1.5 rpm. The biggest original harmonic is 12th order with magnitude is 0.889 rpm which is selected speed harmonic order to reduce. This harmonic order serves as a reference and will be set into the model for minimization.

IV. SIMULATION RESULTS

A. Speed harmonic minimization

In this section, the proposed injected voltage harmonic is generated by the resonant controller in Simulink. To address the issue of torque ripple, current and speed controller loops are designed in the PMSM system is shown in Fig. 3. The

parameters of PMSM are given in Table I. The 12th order electrical frequency speed harmonic is produced, and the reference speed is 10 rpm. The speed and current loops have the same frequency with system sampling frequency.

The simulation results demonstrate the effectiveness of speed harmonic in Fig. 5 (a) and torque ripple in Fig. 5 (b) suppression. By generating an additional voltage harmonic in Fig. 5 (c), both the 12nd-order speed harmonic and torque ripple are significantly reduced.

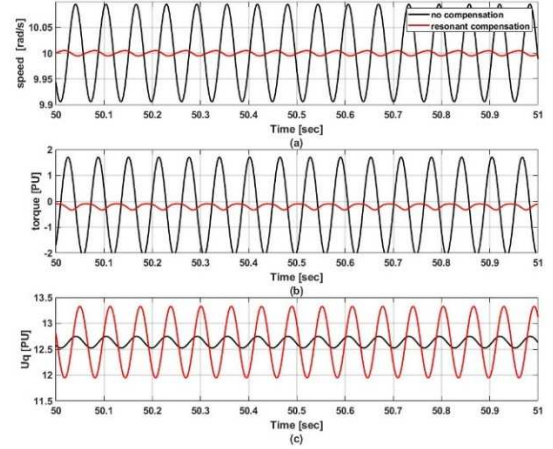


Fig. 5. Simulation results of proposed method. (a) Speed time-domain curve. (b) Rotor torque time-domain curve. (c) voltage time-domain curve

The FFT analysis results are shown in Fig. 6, illustrating that the speed harmonic is suppressed from 0.0938 rad/s to 0.0023 rad/s, while the torque ripple is reduced from 1.8765 per unit to 0.0633 per unit. These results confirm the feasibility of the proposed method, showcasing its strong performance. Due to its simpler design which utilizing a resonant controller for direct voltage harmonic compensation, it offers high reliability and ease of implementation.

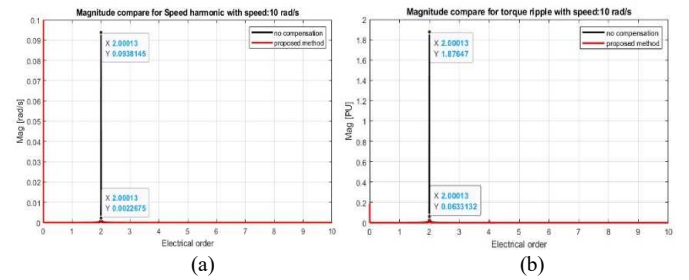


Fig. 6. FFT analysis of speed harmonic and torque ripple in simulation

V. EXPERIMENT ANALYSIS

In this section, the proposed method is implemented in two test benches under with and without load conditions to test and validate speed harmonic and torque ripple minimization performance. The first test bench is a prototype comprising a SPMSM motor with an encoder and a servo drive controlled by a System-on-Chip (SoC)-based controller. It is utilized to

evaluate the performance of speed harmonic minimization which is mentioned in part A and B. The second test bench consists of a motor, gearbox, payload, and other components, as depicted in in Fig. 9. The proposed method is implemented on this test bench to validate its effectiveness in minimizing torque ripple in part C.

A. Speed harmonic minimization at low speed without load condition

The speed harmonic minimization is testing on the test box which is a three-phase SPMSM with five pole-pair. The FOC and motor control algorithms have been implemented in the software, while the measurement and modulation have been done in firmware which are supplied by ABB CRC Sweden. The speed signal is obtained from the encoder.

Before applying the proposed method, the dominant speed harmonic was identified as the 12th electrical order harmonic, which reaches its maximum amplitude at 30 rpm. Fig. 7 (a) compares the speed harmonic with and without compensation in a time-domain. The portion before 253.67s corresponds to operation with compensation, while the portion after 253.67s shows without compensation. The results show that the proposed compensation method significantly reduces the speed harmonic.

As shown in Fig. 7 (b), the total harmonic magnitude without compensation is approximately 2.8 rpm at 30 rpm. The largest harmonic component is the 12th electrical order, with a magnitude of 2.01 rpm. After applying the proposed method, the total harmonic magnitude is reduced to 0.5 rpm, and the 12th-order harmonic decreases to 0.249 rpm.

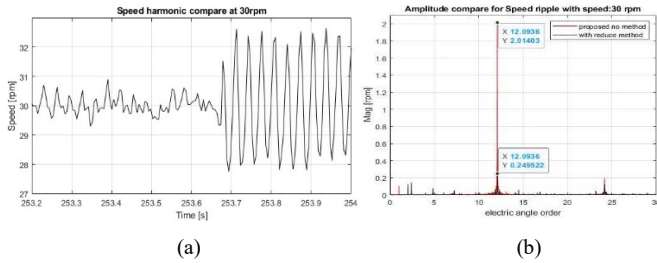


Fig. 7. Comparing with speed harmonic between proposed method and no compensation method. (a) Speed time-domain curve. (b) Speed frequency-domain curve

B. multiply speed harmonics minimization at low speed with load condition

Furthermore, the test is done for multiply speed harmonics compensation by proposed method. Under load condition, 12th and 2nd electrical frequency order harmonic are obtained at 30 rpm. After compensation which the portion is after 79.4s by the proposed method, the magnitude of total speed harmonic is significant reduced in time domain which is shown in Fig. 8.

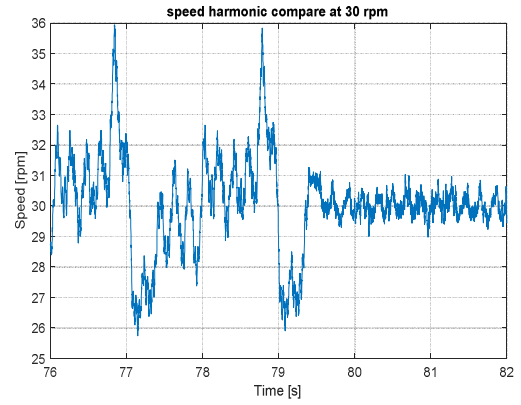


Fig. 8. Speed harmonic minimization on 12th and 2nd electrical frequency at 30 rpm in time-domain

C. Torque ripple minimization at low speed with load condition

The torque ripple minimization verification platform consists of two SPMSM motors with coupled shafts and a torque sensor in Fig. 9. The left SPMSM serves as test machine and the right SPMSM applies load to the test machine. The two machines have different pole-pair numbers, so the harmonics from the load machine can be identified from the test machine measurement. The speed signal and torque signal are measured from torque sensor. The left motor pole pair number is 5.

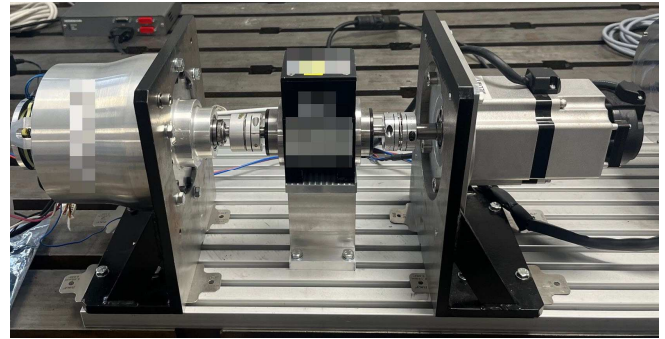


Fig. 9. Experimental setup of PMSM drive system used for torque ripple minimization validation

The comparison results of the FFT analysis for torque (Fig. 10(a)) and speed (Fig. 10(b)) show the harmonic components before and after compensation at 30 rpm. The speed harmonic and torque ripple both have the biggest harmonic at 2nd electrical frequency. The original magnitude of speed harmonic is 1.29 rpm, and the magnitude of torque ripple is 0.014 per unit. After using the proposed compensation method, the magnitude of speed harmonic is reduced to 0.28 rpm and the magnitude of torque ripple is reduced to 0.0018 per unit. The results show that torque ripple is significantly reduced with good performance when speed harmonic is reduced as the situation with load.

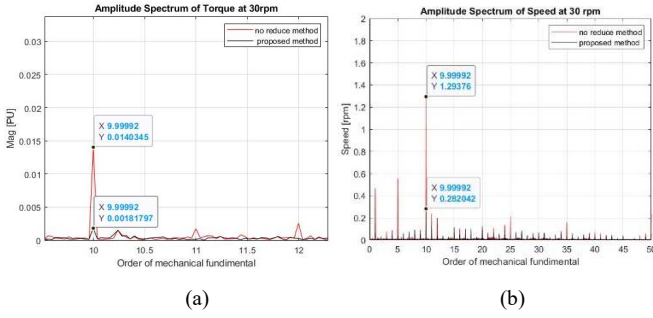


Fig. 10. Speed harmonic and torque ripple comparison between proposed method and no compensation method at 30 rpm. (a) Torque in frequency domain. (b) Speed in frequency domain

VI. CONCLUSION

This paper proposes a voltage harmonic compensation method for minimizing torque ripple in SPMSMs. The voltage harmonic is generated by a resonant controller based on speed harmonics, effectively suppressing both speed harmonics and torque ripple. Due to its simplicity, the method eliminates the need for a current harmonic compensation link, enhancing system stability and robustness. It is easy to implement in robotic applications and operates independently of motor and control parameters. Additionally, it does not require speed or current harmonic extraction, improving the system's dynamic response and accuracy. Furthermore, the method offers flexibility in harmonic frequency compensation, minimizing torque ripple not only for harmonic frequencies of the form $6n$ but also for any n th electrical frequency order.

TABLE I. MOTOR SYSTEM PARAMETERS SET

Parameters	Sign	Value	Unit
The number of harmonic order	n	2	-
Motor pole pair number	p	5	-
Speed PI controller	K_{pv}	2	-
	K_{iv}	2	-
Current PI controller	K_{pd}	10	-
	K_{pq}	10	-
	K_{id}	15	-
	K_{iq}	15	-
Resonant controller	K_{rn}	30	-
	φ_n	pi/3	rad
Rotational mechanical speed	ω_r	10	rad/sec
Sample time	T_s	0.0625	ms
Inductance	L_s	0.0044	H
Permanent magnet rotor flux linkage	ψ_{pm}	0.2538	Wb
Inertia	J	0.2	kg $\cdot$ m ²
Viscous friction	B	0.001	N $\cdot$ s/m ²
Torque load	τ_L	-0.2	N $\cdot$ m

REFERENCES

- [1] M. King, B. Zhu, and S. Tang, "Optimal path planning," *Mobile Robots*, M. S. Rafiq, W. Midgley and T. Steffen, "A Review of the State of the Art of Torque Ripple Minimization Techniques for Permanent Magnet Synchronous Motors," in *IEEE Transactions on Industrial Informatics*, vol. 20, no. 1, pp. 1019-1031, Jan. 2024, DOI: 10.1109/TII.2023.3272689.
- [2] Y. Shakhin, K. Talapiden, N. G. Minh Thao, M. Bagheri and T. Duc Do, "Analysis and Design Optimization of Surface Permanent Magnet Motor to Improve Torque Density and Ripple," 2023 11th International Conference on Power Electronics and ECCE Asia (ICPE 2023 - ECCE Asia), Jeju Island, Korea, Republic of, 2023, pp. 2308-2311, DOI: 10.23919/ICPE2023-ECCEAsia54778.2023.10213949.
- [3] Qu, J., Zhang, P., Jatskevich, J., & Zhang, C. (2023). Torque ripple reduction of permanent magnet synchronous machine drives based on novel speed harmonic control at low-speed operation. *IEEE Transactions on Industrial Electronics*, 70(8), 1-11.
- [4] C. Lai, G. Feng, K. Iyer, K. Mukherjee, and N. C. Kar, "Genetic algorithm based current optimization for torque ripple reduction of interior PMSMs," *IEEE Trans. Ind. Appl.*, vol. 53, no. 5, pp. 4493-4503, Sept. 2017.
- [5] Qu, J., Zhang, P., & Jatskevich, J. (2024). Harmonic current optimization for torque ripple reduction in permanent magnet synchronous machine drives based on torque ripple surrogate model. *IEEE Transactions on Power Electronics*, 1-11. <https://doi.org/10.1109/TPEL.2024.3352588>.
- [6] J. Qu, J. Jatskevich, C. Zhang and S. Zhang, "Torque ripple reduction method for permanent magnet synchronous machine drives with novel harmonic current control," *IEEE Trans. Energy Convers*, vol. 36, no. 3, pp. 2502-2513, Sept. 2021.
- [7] D. Flieller, N. K. Nguyen, P. Wira, G. Sturtzer, D. O. Abdeslam, and J. Merckle, "A self-learning solution for torque ripple reduction for nonsinusoidal permanent-magnet motor drives based on artificial neural networks," *IEEE Trans. Ind. Electron.*, vol. 61, no. 2, pp. 655-666, Feb. 2014.
- [8] G. Feng, C. Lai, and N. C. Kar, "A closed-loop fuzzy-logic-based current controller for PMSM torque ripple minimization using the magnitude of speed harmonic as the feedback control signal," *IEEE Trans. Ind. Electron.*, vol. 64, no. 4, pp. 2642-2653, Apr. 2017.
- [9] C. Xia, B. Ji and Y. Yan, "Smooth Speed Control for Low-Speed High-Torque Permanent-Magnet Synchronous Motor Using Proportional-Integral-Resonant Controller," in *IEEE Transactions on Industrial Electronics*, vol. 62, no. 4, pp. 2123-2134, April 2015, DOI: 10.1109/TIE.2014.2354593.
- [10] Jian Gao, Xuan Wu, etc. "Torque ripple minimisation of permanent magnet synchronous motor using a new proportional resonant controller," *IET Power Electronics*, 2017, Vol. 10, Iss. 2, pp. 208-214. DOI: 10.1049/iet-pel.2015.0946.
- [11] E. A. Ignatev and O. V. Nos, "Torque Ripple Reduction of Permanent Magnet Synchronous Motor Using Proportional-Integral-Resonant Controller with Delay of Control Response Compensation," *2021 IEEE 22nd International Conference of Young Professionals in Electron Devices and Materials (EDM)*, Souza, the Altai Republic, Russia, 2021, pp. 431-435, DOI: 10.1109/EDM52169.2021.9507607.
- [12] Z. Wu, Q. Liang, S. Zhang, J. Liang and X. Yang, "Amplitude-phase based optimal voltage harmonic injection for speed harmonic minimization in SPMSM," *IEEE Trans. Power Electron.*, vol. 38, no. 6, pp. 7494-7503, June 2023.